

The layer illusion of motor cortical beta rhythms

Björg Elisabeth Kilavik¹, Laura López-Galdo^{1,2}, Simon Nougaret¹, Demian Battaglia²

¹Institut de Neurosciences de la Timone (INT), UMR 7289, CNRS, Aix-Marseille Université, Marseille, France

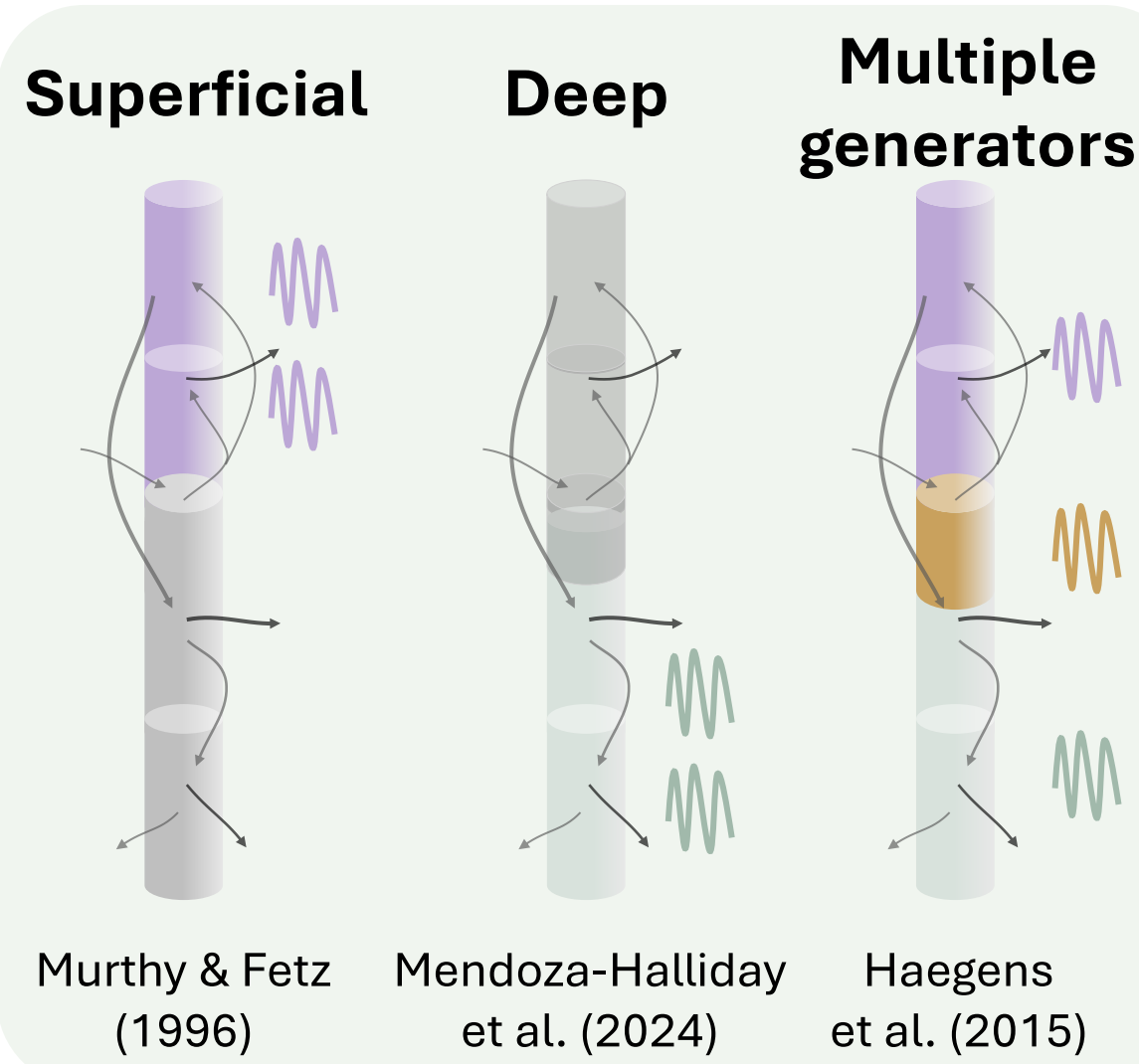
²University of Strasbourg Institute for Advanced Studies (USIAS), F-67000 Strasbourg, France

email: Bjorg.kilavik@univ-amu.fr

1. Introduction

Cortical layer sources of beta

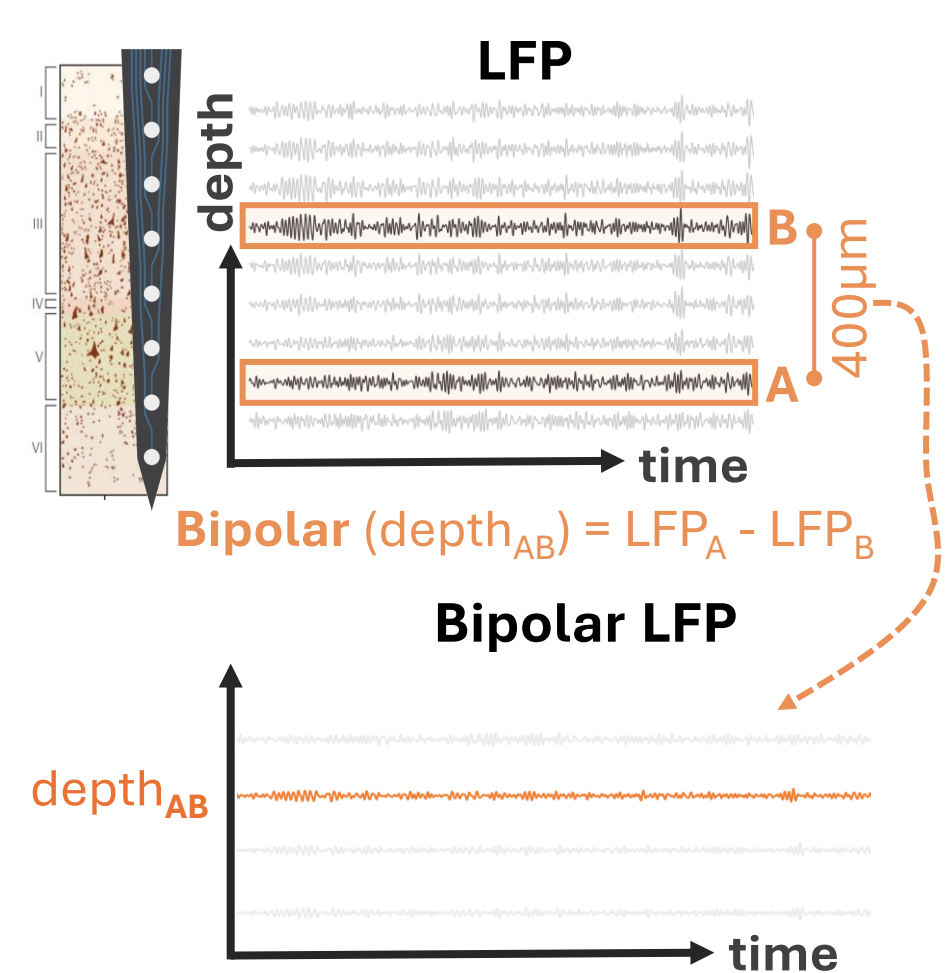
- Beta rhythms are **dominant** in motor cortex
- Functional roles** and **cortical layer sources** still disputed:
 - Superficial* and involved in *sensorimotor integration/attention?*
 - Deep* and involved in *top-down feedback?*
 - Multiple *independent generators* with potentially *different roles?*



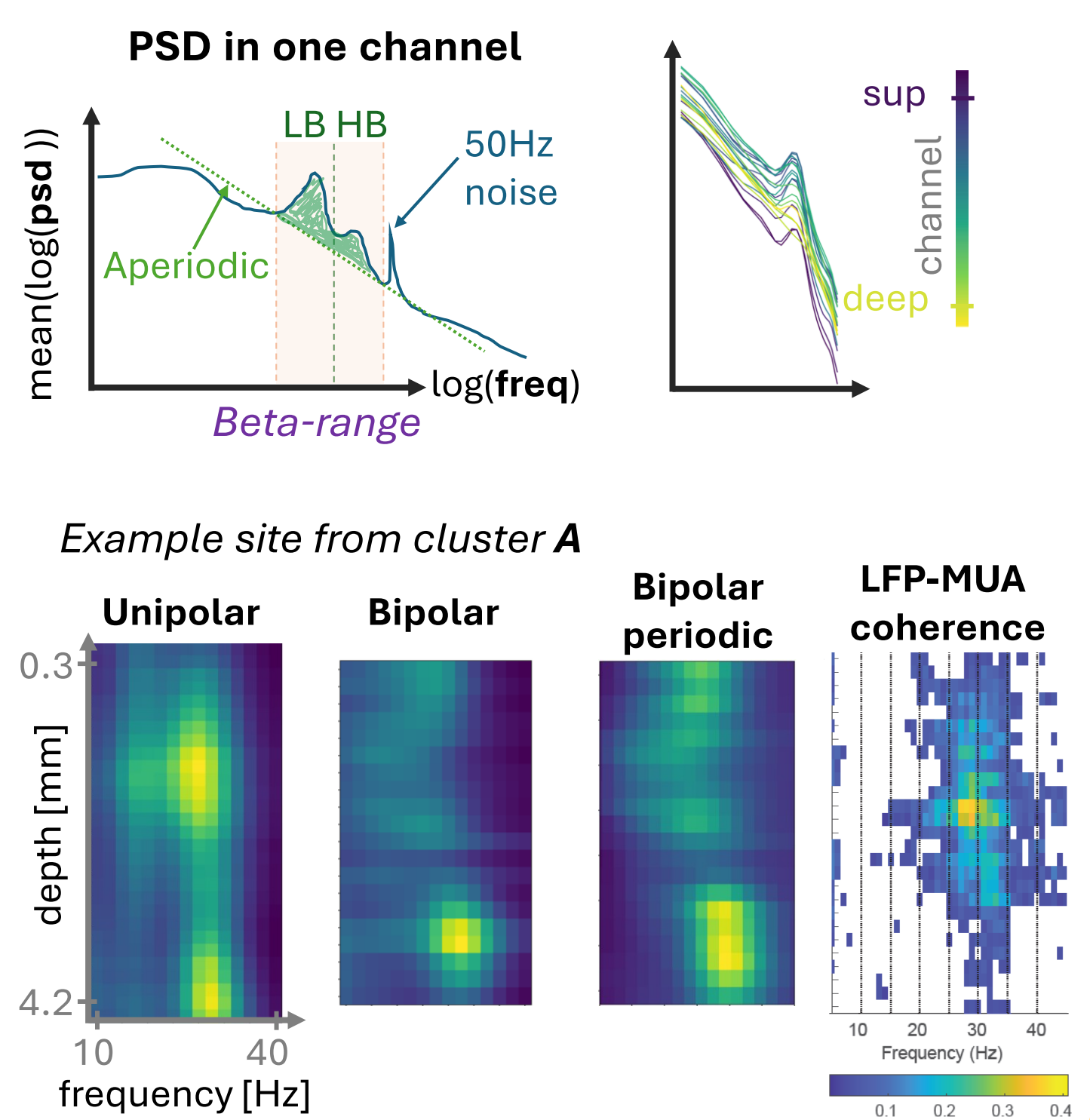
3. Methods

Pre-processing

Local re-referencing



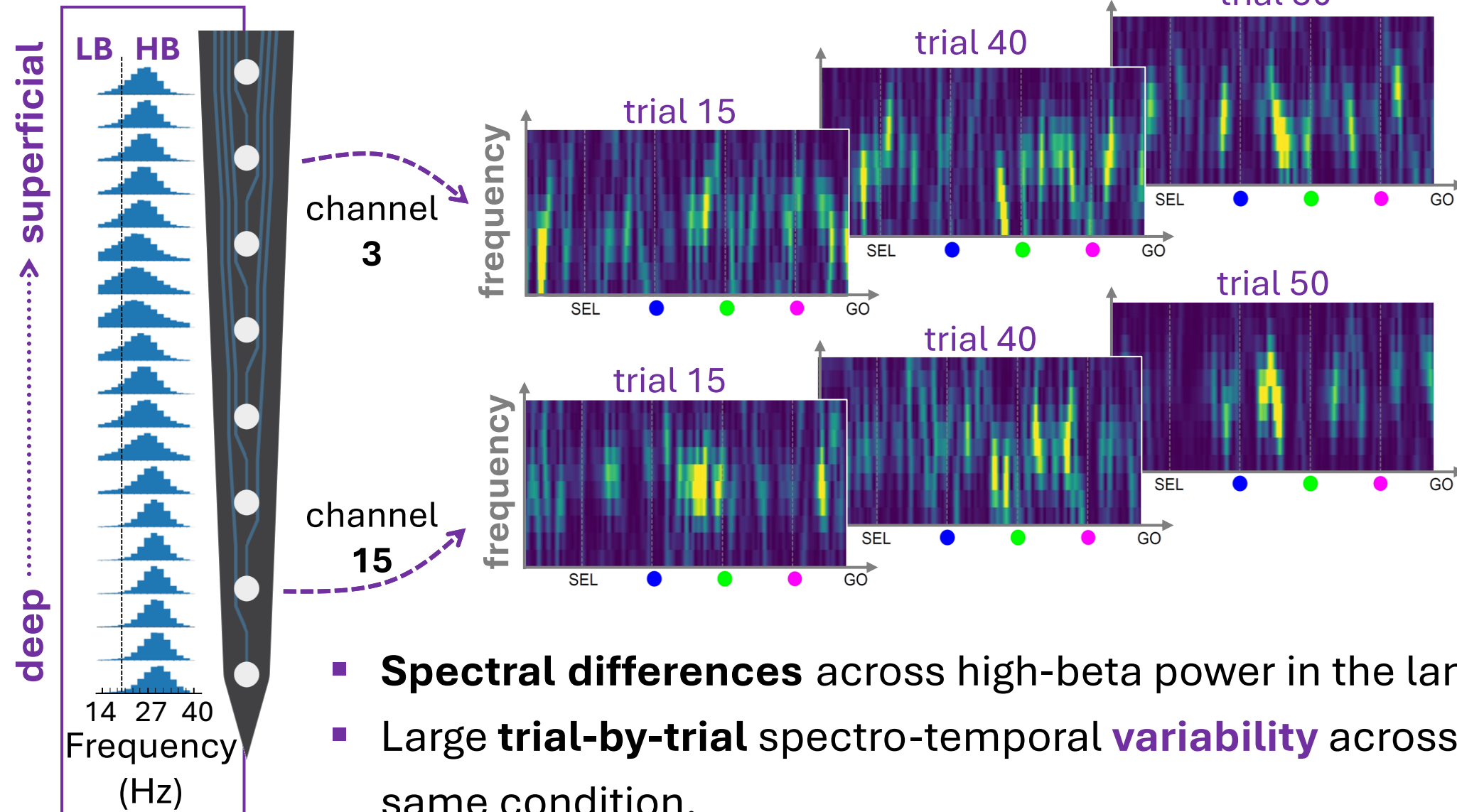
Periodic-only beta power



- Using **bipolar derivation**, we minimize the effects of volume conduction, localizing the LFP signal.
- We estimate and remove the aperiodic (1/f) part of the power in each channel independently.

4. Beta Variability

Spatio-Spectral Variability

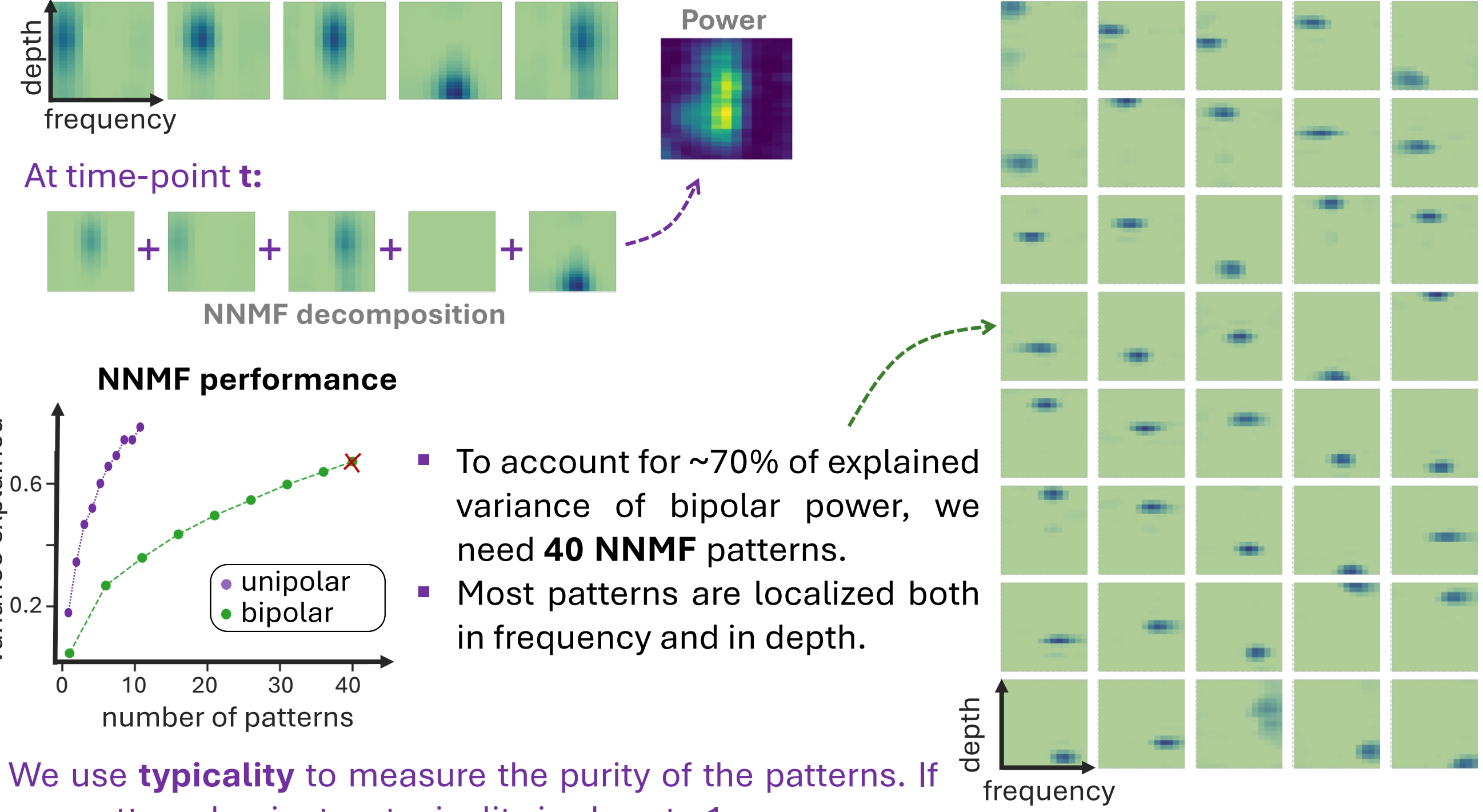


- Spectral differences** across high-beta power in the laminar dimension.
- Large **trial-by-trial** spectro-temporal **variability** across trials of the same condition.

Spectro-Laminar Localization

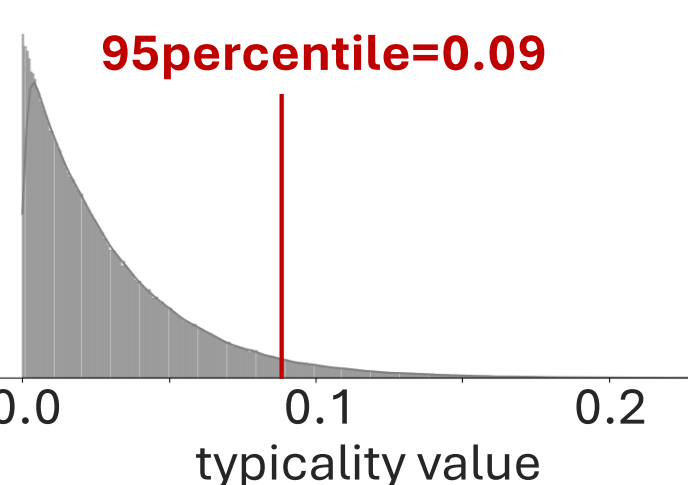
- Variability occurs across laminae and frequency and its complete description is high-dimensional (e.g. D x F). **Need for dimensional reduction!**
- We applied **Non-Negative Matrix Factorization (NNMF)** in single trial bipolar power.

Pattern decomposition example (k=5)

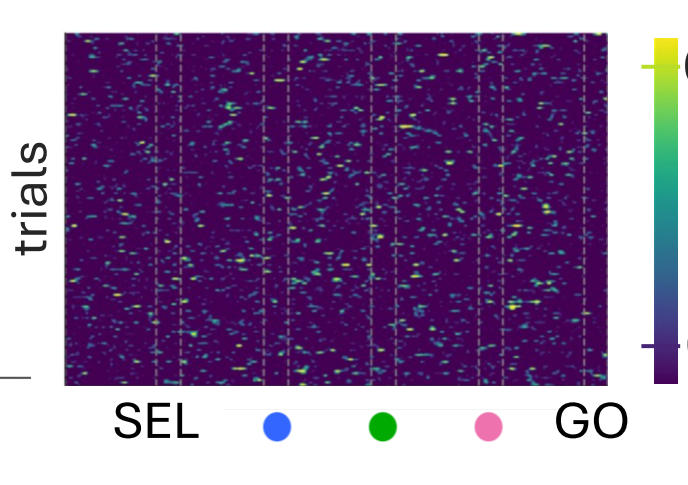


We use **typicality** to measure the purity of the patterns. If one pattern dominates, typicality is close to 1.

Typicality distribution



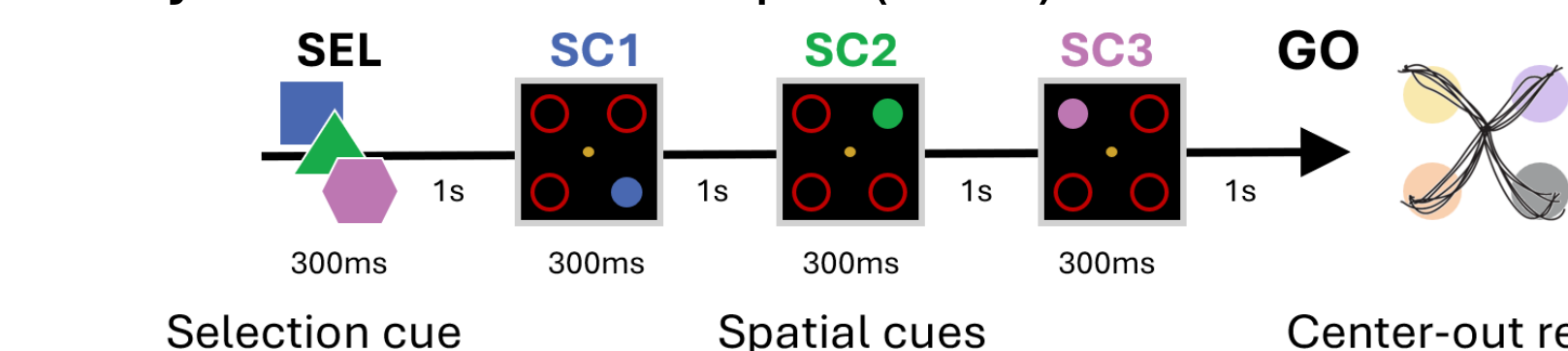
Strongest pattern value



- It's typical to be **atypical!**
- Despite the precise localization of individual patterns, patterns are **never alone**. At every time-point there's a **mixture of patterns**, not one dominating.

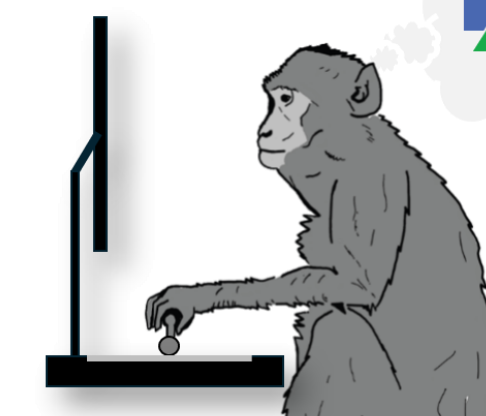
2. Behavioral paradigm

Visuomotor task - Temporally predictive and fixed order delayed-match-to-sample (DMS)

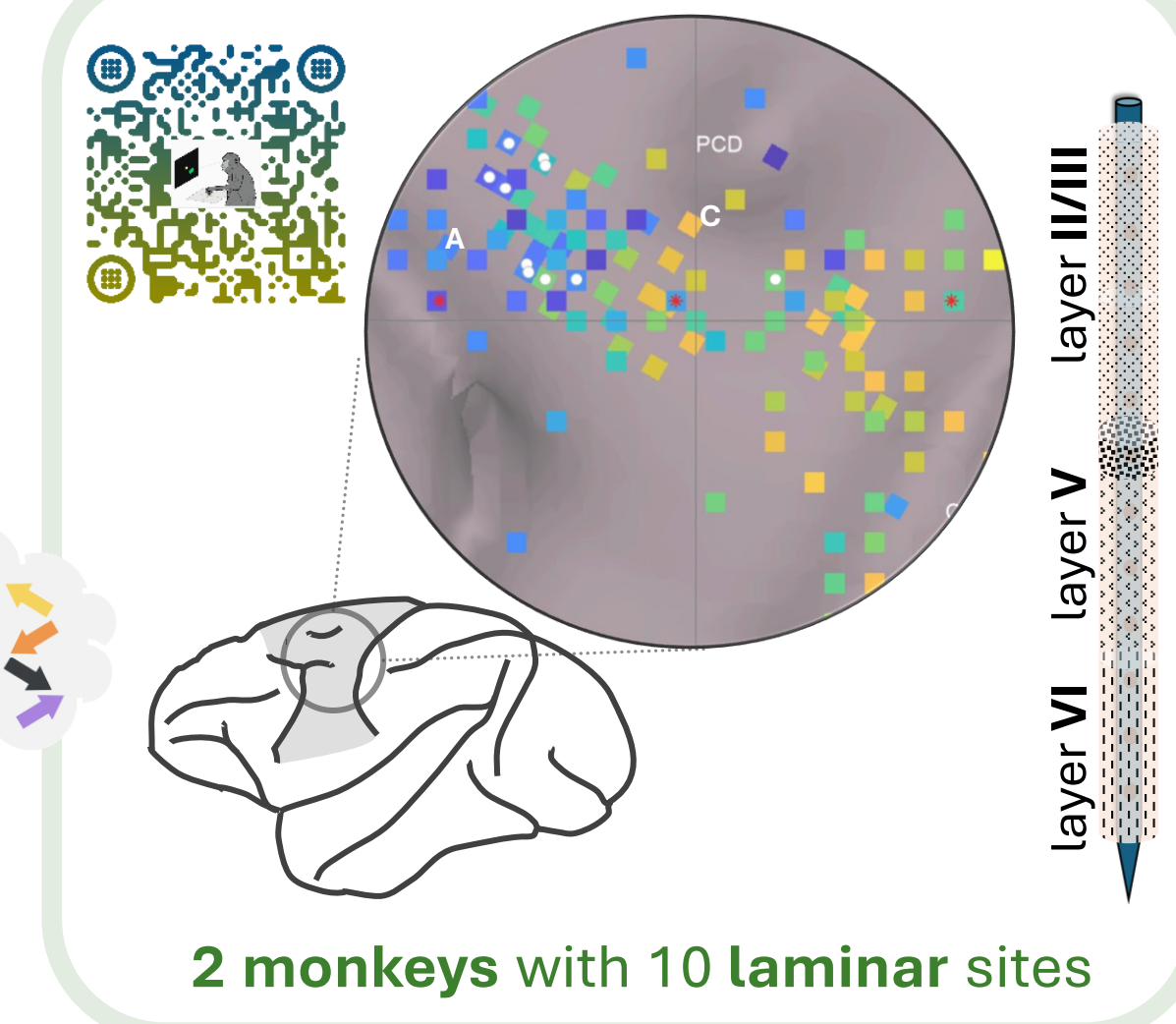


Monkeys performed a reach in the **direction** of the spatial cue (SC) that matched the **color** of the selection cue (SEL).

- SEL**: **blue, green or pink**.
- SCs**: 4 possible directions (random).
- The **color order** of the SCs is **fixed**.
- The monkey should stabilize the cursor in the center until the **GO**.



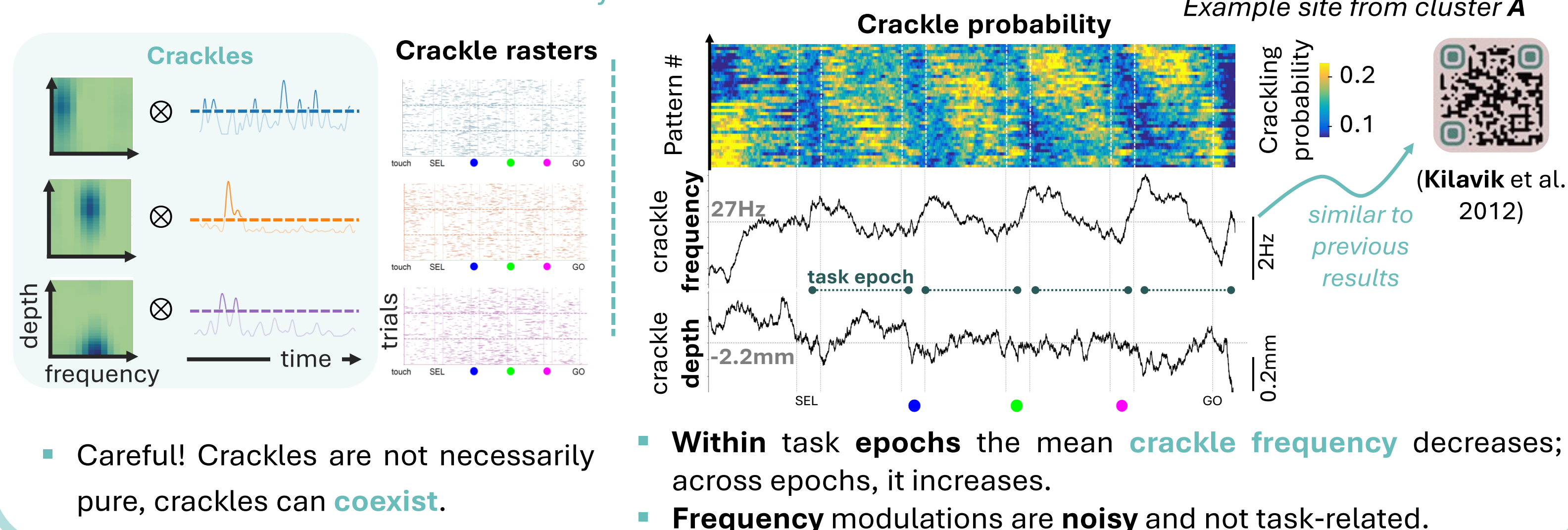
Dataset



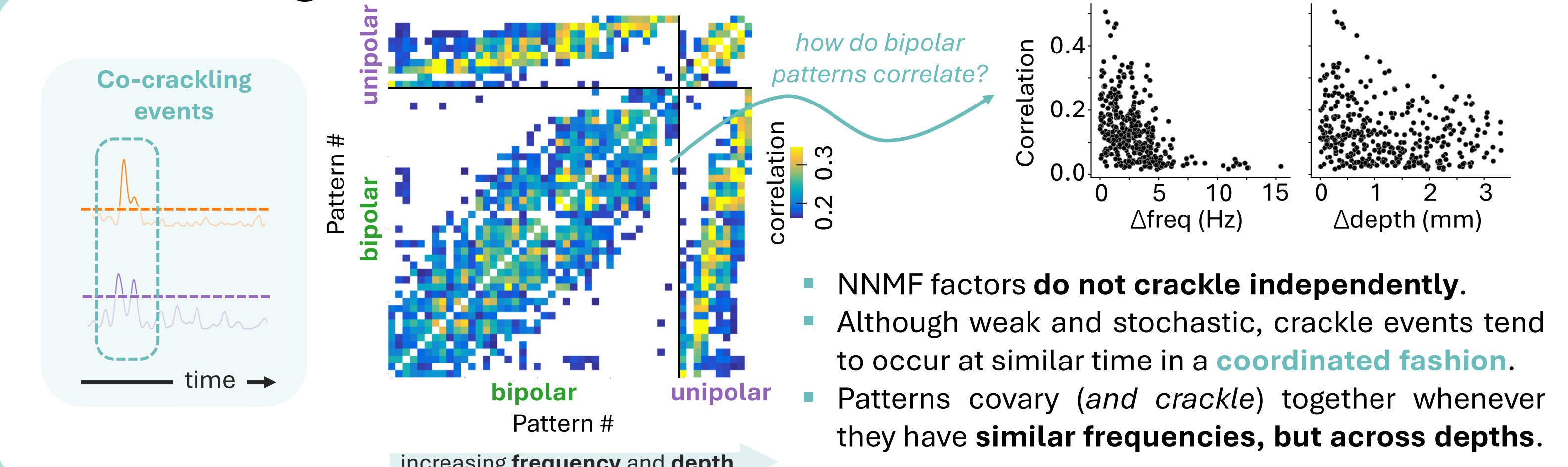
5. Beta events

Crackles

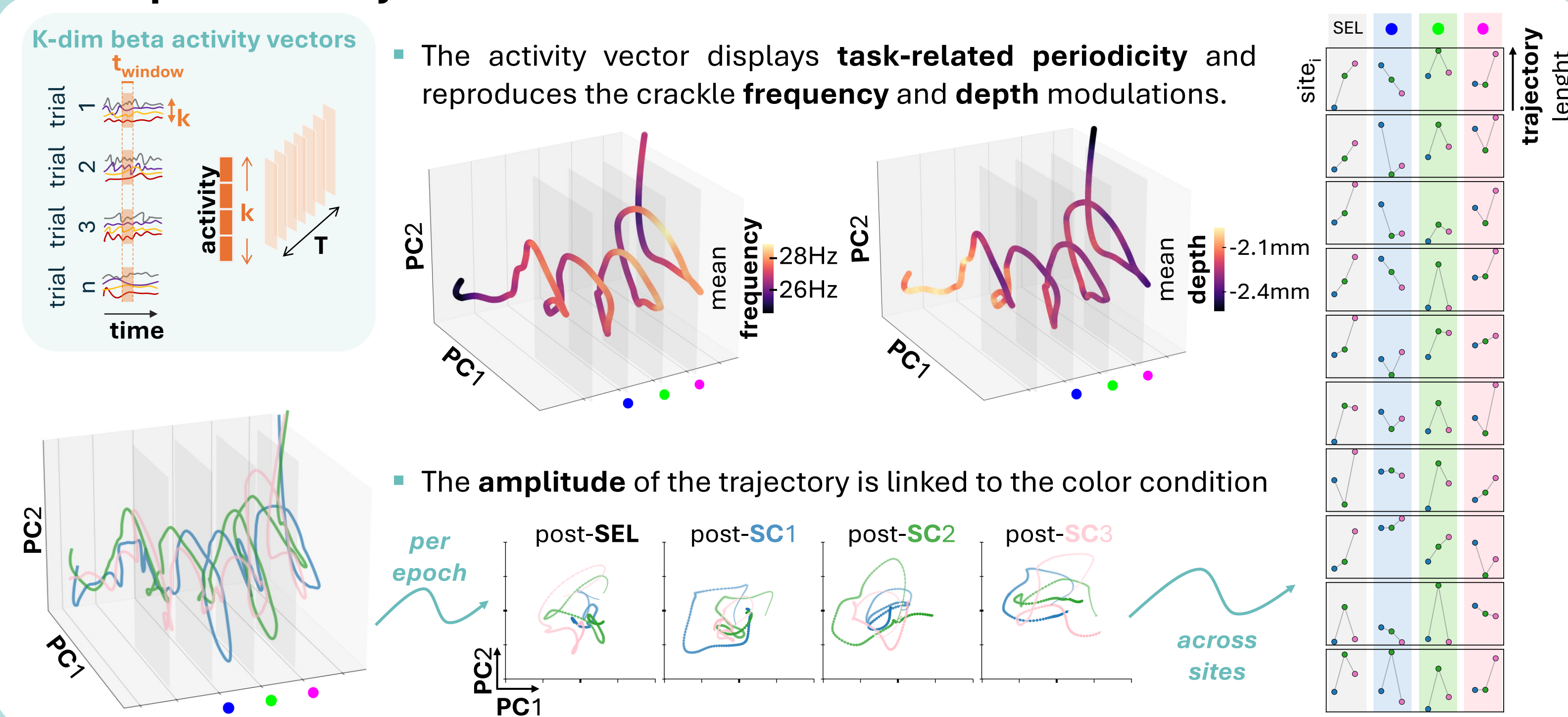
Events that are **relatively strong**, i.e. they are above a relative (not absolute) threshold. It captures transient increases of a factor's activity.



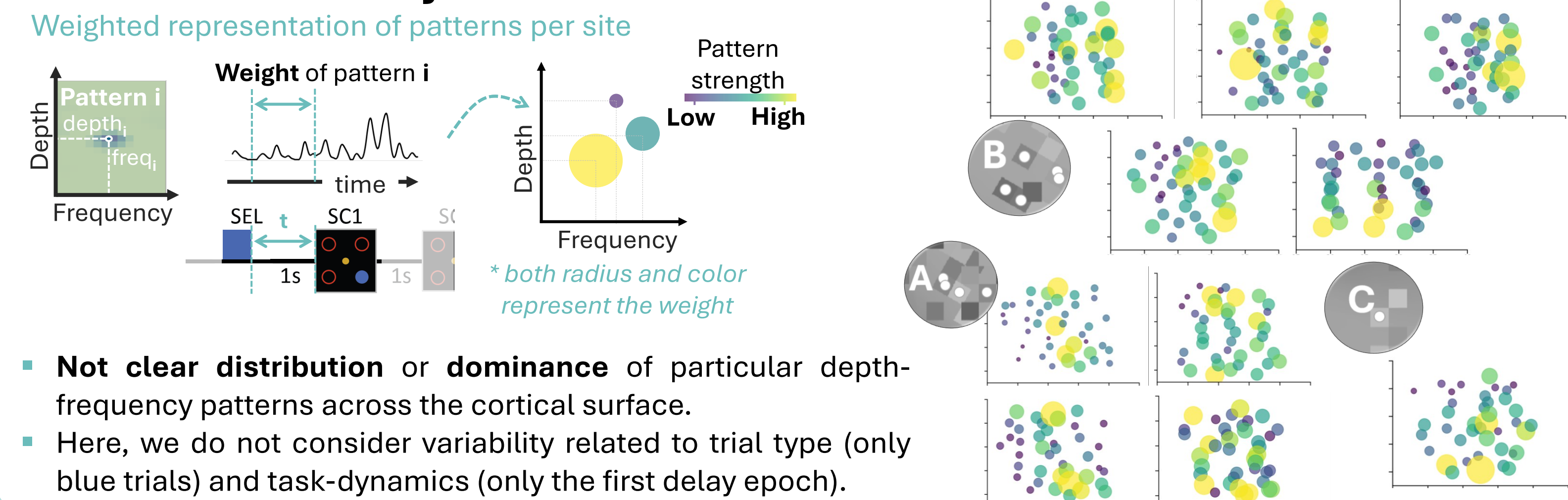
Co-crackling



Beta pattern trajectories



Cortical Variability of Patterns



6. Conclusions

- Spectro-laminar patterns are always mixed, suggesting **no clear depth localization** of individual beta events.
- Individual neurons can span multiple layers, pointing towards **multi-laminar cogeneration** of beta events.
- Relative contributions of spectro-laminar components are highly **diverse** across the motor cortical sheet.
- When taken together, patterns display **task-related dynamics**.

Co-generation of beta across layers

